

Module 8: Implicit Curves and Conics – Prequiz

Name:

Date:

Four short questions. Read each equation geometrically.

1. Identify the conic: $\frac{x^2}{16} + \frac{y^2}{9} = 1$.

2. For $(x - 2)^2 + (y + 1)^2 = 25$, state the center and radius.

3. For $x^2 = -12y$, state the direction in which the parabola opens.

4. For $\frac{x^2}{9} - \frac{y^2}{4} = 1$, state the vertices.

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Three questions. Read each equation geometrically and show enough work to support your answers.

1. Reading conics from standard form.

For each equation, identify the conic and give the requested information.

Equation	Conic	Requested information
$(x - 1)^2 + (y + 2)^2 = 16$		Center and radius
$(x - 2)^2 = 8(y + 1)$		Vertex and opening direction
$\frac{(x + 1)^2}{16} - \frac{(y - 2)^2}{9} = 1$		Center and vertices

2. An ellipse.

For

$$\frac{(x - 1)^2}{16} + \frac{(y + 2)^2}{9} = 1,$$

state the center, vertices, covertices, and foci exactly.

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3. Graphing the ellipse.

Graph

$$\frac{(x-1)^2}{16} + \frac{(y+2)^2}{9} = 1.$$

Label the center, both vertices, both covertices, and both foci. Your graph should agree with the information you found in Question 2.

